

Counterexamples to unimodality of the connected set polynomial

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Abstract

For a finite graph G , let $C(G; x) = \sum_{k \geq 1} c_k(G) x^k$, where $c_k(G)$ counts the k -vertex subsets inducing a connected subgraph. A question of Mol asks whether $C(G; x)$ is unimodal whenever G is connected and $|E(G)| \geq |V(G)|$. For every order $n \geq 8$, we give four explicit pairwise nonisomorphic counterexamples, with edge counts $n+1, n+2, n+2, n+3$. Their polynomials follow from a common rooted-block decomposition and exhibit the fixed local pattern 10, 11, 10. We also enumerate all connected graphs of order eight satisfying the edge condition and prove that exactly four isomorphism classes are counterexamples; they are precisely the first members of our four families. No eligible graph of smaller order is a counterexample. Thus the minimum counterexample order is eight.

1 Introduction

The connected set polynomial of a finite simple graph G is

$$C(G; x) = \sum_{k=1}^{|V(G)|} c_k(G) x^k, \quad (1.1)$$

where $c_k(G)$ is the number of k -element sets whose induced subgraph is connected. This polynomial is the one-component part of the induced-subgraph component polynomial introduced by Tittmann, Averbouch and Makowsky [3], and its coefficients are closely related to node reliability.

Mol asked whether the coefficient sequence is always unimodal for a connected n -vertex graph with at least n edges; the same paragraph reports a computational check through order eight. The statement appears in the conclusion chapter, under “Connected Set Polynomial,” of the 2016 thesis cited below [2]. The theorem below answers the question negatively and determines the complete minimum-order boundary.

Theorem 1. *For every $n \geq 8$, there are four pairwise nonisomorphic connected n -vertex graphs with at least n edges whose connected set polynomials are not unimodal. Their edge counts are*

$$n+1, \quad n+2, \quad n+2, \quad n+3.$$

Among all connected eight-vertex graphs with at least eight edges, exactly four isomorphism classes have a nonunimodal connected set polynomial, and they are the four order-eight graphs in these families. No eligible graph of order at most seven is a counterexample.

2 Four rooted-block families

Fix $d \geq 2$. Join rooted graphs L and R by a path of length d , identifying its endpoints with the roots $u \in L$ and $v \in R$.

For the left block choose either $D = K_4 - e$, rooted at a vertex of degree three, or $K = K_4$. For the right block choose either $S = K_{1,2}$, rooted at its centre, or $T = K_3$. Denote the four resulting graphs by

$$G_d^{D,S}, \quad G_d^{D,T}, \quad G_d^{K,S}, \quad G_d^{K,T}.$$

Let ε be one when the left block is K and zero otherwise, and let η be one when the right block is T and zero otherwise. Every graph has

$$n = d + 6, \quad m = n + 1 + \varepsilon + \eta.$$

The four graphs are pairwise nonisomorphic. Their block decompositions recover whether the left nontrivial block is $K_4 - e$ or K_4 , and whether the right endpoint lies in a triangle.

3 Rooted decomposition

For a rooted graph (H, r) , let $A_H(x)$ count the nonempty connected induced vertex sets containing r . In both left blocks the root is adjacent to all three other vertices, so

$$A_L(x) = x(1 + x)^3.$$

Similarly, both right blocks satisfy

$$A_R(x) = x(1 + x)^2.$$

The sum of the two ordinary connected set polynomials is

$$C(L; x) + C(R; x) = 7x + (7 + \varepsilon + \eta)x^2 + 5x^3 + x^4.$$

We partition a connected induced set according to whether it lies in an endpoint block, lies wholly inside the $d - 1$ internal path vertices, meets one endpoint block and an initial path segment, or meets both endpoint blocks. These classes are disjoint and exhaustive. Consequently,

$$\begin{aligned} C(G_d; x) &= 7x + (7 + \varepsilon + \eta)x^2 + 5x^3 + x^4 + \sum_{k=1}^{d-1} (d - k)x^k \\ &\quad + (A_L(x) + A_R(x))(x + x^2 + \cdots + x^{d-1}) + A_L(x)A_R(x)x^{d-1}. \end{aligned} \quad (3.1)$$

Expanding (3.1) yields

$$\begin{aligned} C(G_d; x) &= nx + (n + 1 + \varepsilon + \eta)x^2 + (n + 3)x^3 \\ &\quad + \sum_{k=4}^{d+2} (n + 6 - k)x^k + 11x^{d+3} + 10x^{d+4} + 5x^{d+5} + x^{d+6}. \end{aligned} \quad (3.2)$$

The sequence decreases after $c_3 = n + 3$, reaches $c_{d+2} = 10$, and then rises to $c_{d+3} = 11$. Hence all four polynomials are nonunimodal. Since $d = n - 6$, this proves the infinite-family part of Theorem 1.

4 The minimum-order classification

At $d = 2$, equation (3.2) becomes

$$(8, 9 + \varepsilon + \eta, 11, 10, 11, 10, 5, 1).$$

We checked Brendan McKay’s complete `graph8c.g6` catalogue of 11,117 nonisomorphic connected graphs of order eight. Exactly 11,094 have at least eight edges, and exactly four of these have a nonunimodal connected set polynomial. This is a complete catalogue scan, not a generated sample. Their graph6 encodings are

```
G?B@vG
G?`crg
G?aJeW
GCQRTg
```

Isomorphism tests identify them with the four $d = 2$ constructions above. An independent scan of the NetworkX graph atlas finds no eligible counterexample of order at most seven. The connected-set coefficients were computed by two independent routines: direct subset connectivity and boundary expansion. They agree on every tested graph. This proves the finite classification in Theorem 1.

The catalogue is available from <https://users.cecs.anu.edu.au/~bdm/data/graph8c.g6>. Its expected SHA-256 digest is

0002354F1AB3344A2706626A037AD15367BF23A2163AA68F552C3A169CA9A036.

The verification used Python 3 and NetworkX 3.6.1. The supplied verification directory records the complete command and the logical boundary between the formula checks, the atlas scan and the full order-eight catalogue scan.

5 Scope

The theorem does not classify all graphs with nonunimodal connected set polynomials. In particular, we do not claim that four is the number of counterexamples in orders greater than eight, or that no counterexample with exactly n edges exists. The result identifies four explicit infinite families and completely settles only the minimum-order boundary.

References

- [1] A. Dossou-Olory, *Graphs and unicyclic graphs with extremal number of connected induced subgraphs*, arXiv:1812.02422.
- [2] L. Mol, *On Connectedness and Graph Polynomials*, PhD thesis, Dalhousie University, 2016, conclusion chapter, “Connected Set Polynomial” subsection; [stable thesis record](#) and [PDF](#).
- [3] P. Tittmann, I. Averbouch and J. A. Makowsky, *The Enumeration of Vertex Induced Subgraphs with respect to the Number of Components*, European Journal of Combinatorics 32 (2011), arXiv:0812.4147.

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